Pulkit Girdhar

B.Tech. (Electrical Engineering), Indian Institute of Technology Roorkee
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Research Interests

Machine learning · Scientific ML & surrogate modelling · Uncertainty quantification · Quantum computing

Education

Indian Institute of Technology Roorkee — CGPA: **9.00**/10
B.Tech. (Electrical Engineering) — *2024 – 2028 (expected)*

B.G.N. Adarsh Public School, Bathinda — 90.60%
Class XII (CBSE) — *2024*

St. Xavier's School, Bathinda — 92.00%
Class X (CBSE) — *2022*

Research Experience

Context Compression & Adaptation in Prior-Data Fitted Networks — May 2026 – July 2026
Trust Lab, IIT Bombay — Advisor: Prof. Pratik Jawanpuria (CMInDS)

- Studied **how much in-context data TabPFN needs**, and whether LoRA-specialized models **transfer across synthetic data-generating priors**.
- Built a **transformer attention-analysis pipeline** extracting TabPFN's internal attention distributions to rank training samples by informativeness for compressed inference.
- Fine-tuned LoRA adapters on frozen TabPFN across synthetic priors from four data-generating families; found **cross-prior transfer degrades as Wasserstein/MMD distance between priors grows**.
- Related prior geometry to adapter geometry via adapter-similarity metrics, tested it under cross-family ablations, and caught and corrected an undertraining artifact that inflated the effect.

GP Surrogates for EV-Charging Feasibility on Power Grids — February 2026 – April 2026
Electrical Engineering, IIT Roorkee — Advisor: Prof. Parikshit Pareek

- Replaced repeated **AC power-flow solves** with **GPU-accelerated Gaussian Process surrogates** (BoTorch, CUDA) for rapid EV-charging feasibility screening.
- Generated Julia AC power-flow datasets across loading conditions; surrogates reached $\sim \mathbf{10^{-5}}$ p.u. voltage and $\sim \mathbf{10^{-4}}$ rad angle error.
- Used **GP predictive uncertainty** (squared-exponential kernel) for voltage-constraint screening, delivered through an interactive feasibility-analysis tool.

Projects

Variational Quantum Classifiers — December 2025 – January 2026
Quantum Computing Group, IIT Roorkee

- Built **PennyLane VQCs** (data re-uploading ansatz, circular CNOT entanglement) with class-weighted loss, early stopping on validation AUC, stratified splits and train-only scaling.
- Benchmarked vs. classical baselines: on credit-card fraud (100k rows, 4 qubits) VQC ROC-AUC **0.959** matched Logistic Regression (0.959) but trailed Random Forest (0.978); same ordering on breast cancer (0.955 vs. 0.965/0.969).

Hydrogen Orbital Solver — December 2024 – February 2025
Advisor: Prof. Abhiraj Sharma, Civil Engineering, IIT Roorkee

- Solved hydrogen-atom wavefunctions from the Schrödinger equation with **Legendre, associated Legendre and Laguerre recursions implemented from scratch**; 3D orbital visualizations across (n, l, m).

Positions of Responsibility & Activities

Member — FuMES Special Interest Group: lead sessions and wrote handouts.
Executive Member — Quantum Computing Group, IIT Roorkee: mentor incoming members.
Member, Technical Council — Vivekananda Bhawan, IIT Roorkee.

Skills

Languages: Python, Julia, C++, MATLAB, SQL
ML & Scientific: PyTorch, BoTorch, PennyLane, scikit-learn, NumPy, SciPy, Pandas
Research Tooling: Git, CUDA, LaTeX, Matplotlib
Relevant Coursework: Numerical Methods, Probability Methods, Data Science, Signals & Systems

References

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Prof. Parikshit Pareek
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